

NLP MINI PROJECT • UNDERGRADUATE EVALUATION

# NEWS ARTICLE CATEGORIZATION

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*Using Natural Language Processing*

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# News Article Categorization

## *Problem, Objectives & Categories*

### THE PROBLEM

News platforms publish a huge volume of articles every day. Sorting each one into the right section by hand is slow, repetitive, and inconsistent — this project automates that sorting step using NLP and machine learning.

### OBJECTIVES

- 1 Clean and normalise raw article text with an NLP pipeline
- 2 Convert text into numerical features using TF-IDF
- 3 Train and evaluate a Logistic Regression classifier
- 4 Deliver a simple, demo-ready web app with real, honest metrics

### FIVE CATEGORIES

**SPORTS**

**TECHNOLOGY**

**BUSINESS**

**POLITICS**

**ENTERTAINMENT**

# Methodology

*NLP Preprocessing → TF-IDF → Logistic Regression*



## PREPROCESSING PIPELINE (STEP-BY-STEP)

**1**

**Lowercase**

*Normalise casing*

**2**

**Remove Punctuation & Numbers**

*Keep only letters*

**3**

**Tokenization**

*Split into words*

**4**

**Stop-word Removal**

*Drop "the, is, and"*

**5**

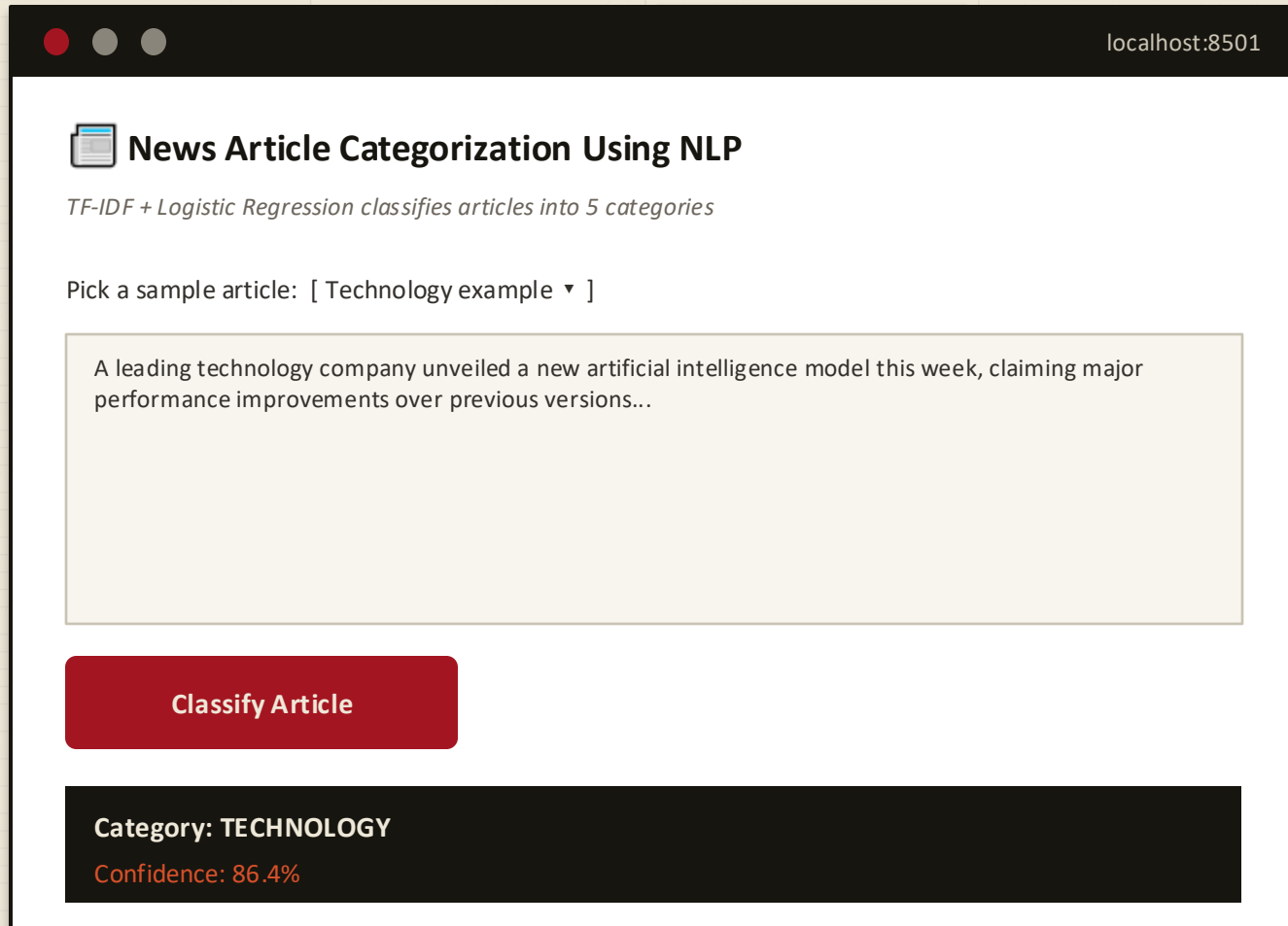
**Lemmatization**

*Reduce to base form*

*TF-IDF weights words by importance across articles; Logistic Regression is simple, fast, and works well on sparse text features — ideal for a viva-explainable pipeline.*

# System & Application

*A simple Streamlit interface demonstrates the full pipeline live*



## HOW IT WORKS

1

### Paste or pick an article

User enters text or selects one of five ready-made examples.

2

### Clean the text

The same `preprocess_text()` function used in training runs on the input.

3

### Vectorize with TF-IDF

The saved vectorizer transforms the cleaned text into features.

4

### Predict with Logistic Regression

The saved model returns a category and class probabilities.

5

### Display result

Category and confidence are shown; low-signal input is handled gracefully.

# Results & Performance

Real metrics from the trained model — nothing hardcoded

95.8%

ACCURACY

95.8%

PRECISION

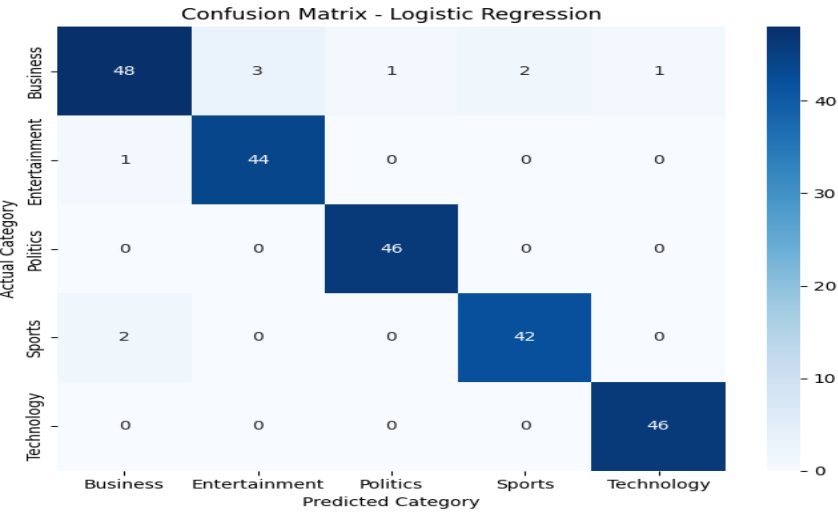
96.1%

RECALL

95.9%

F1-SCORE

## CONFUSION MATRIX



## MODEL COMPARISON



Logistic Regression remains the primary model (fast, interpretable, strong on sparse TF-IDF features). Naive Bayes is shown only as a simple optional comparison, per project scope.

Evaluated on an 80/20 split — 944 train / 236 test articles, random\_state = 42

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# THANK YOU

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*Questions & Discussion*